AHMED MEJRI

HPC SYSTEMS ENGINEER

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HPC Systems Engineer with strong hands-on experience diagnosing HPC application, scheduler, and storage issues, analyzing job failures and performance bottlenecks, specializing in **K8S cluster**, **GPU-based workloads** management, including **NVIDIA GPU** utilization and Multi-Instance **GPU (MIG)** configuration, **PBS Professional** and **Slurm scheduler** administration and optimization, proactive cluster health checks, and monitoring.
Proven expertise in deploying, configuring, and troubleshooting enterprise-scale HPC clusters using OpenHPC and Warewulf provisioning systems, with additional experience in Kubernetes cluster operations for **AI/ML workloads**, including resource allocation, workload scheduling, and GPU-aware workload orchestration, as well as deploying and validating LLM inference services with vLLM on NVIDIA H200 GPUs.
Skilled in parallel computing (**MPI, OpenMP**), InfiniBand/Ethernet networking troubleshooting, Linux administration, storage systems (**Lustre, NFS, GPFS, BeeGFS**), and **Python/Bash automation**, with a strong background in **root cause analysis**, performance optimization, and SOP/runbook documentation across multi-disciplinary hardware and software stacks. **PBS Professional** Certified Associate Specialist with demonstrated ability to support customers running some of the world's most powerful supercomputer clusters. Available for remote HPC Systems Engineer opportunities.

CORE TECHNICAL SKILLS

- **HPC Cluster Management:** PBS Professional, SLURM Workload Manager, OpenHPC, Warewulf provisioning
- **GPU & AI Infrastructure:** NVIDIA GPU management, Multi-Instance GPU (MIG) configuration, GPU workload scheduling and utilization monitoring, LLM inference serving with vLLM (OpenAI-compatible APIs, KV-cache and context-length sizing)
- **Kubernetes Platforms:** K8s cluster operations, resource management, workload scheduling, and cluster health monitoring
- **Parallel Computing:** MPI, OpenMP
- **Diagnostics & Monitoring:** Job failure analysis, performance bottleneck troubleshooting, system log analysis, proactive cluster health checks
- **Storage Systems:** Lustre, NFS, GPFS, BeeGFS
- **Networking:** InfiniBand, RDMA and Ethernet troubleshooting (NVIDIA InfiniBand Essentials certified), Kubernetes networking components (MetalLB, Ingress, Envoy Gateway)
- **Linux Administration & Scripting:** Linux system administration, Python, Bash automation
- **High Availability & Clustering:** Corosync/Pacemaker HA clustering, Kubernetes high availability concepts (control plane / worker resilience), service failover strategies in containerized environments
- **Cloud:** AWS Cloud Essentials, Cloud Infrastructure Technologies
- **Documentation:** SOPs, runbooks, and technical knowledge base maintenance

EXPERIENCE

HPC SYSTEMS ENGINEER, HARRINGTON HPC MICROSYSTEMS
01/2026 – PRESENT

- Provide technical support, administration, and operational management for a

GPU-based High-Performance Computing (HPC) cluster supporting AI, machine learning, and scientific research workloads across multiple environments, including healthcare, education, and research.
- Onboard new users and projects by provisioning HPC and Kubernetes resources, configuring Resource Quotas, storage classes, namespaces, and access permissions, and supporting workload deployment in accordance with operational and security best practices.
- Configure and support NVIDIA Multi-Instance GPU (MIG) profiles to enable GPU partitioning and optimize resource sharing across multiple workloads and users.
- Manage Ingress controllers and routing rules to ensure secure and reliable access to services running on the HPC platform.
- Monitor and maintain the health and performance of the HPC infrastructure, including compute nodes, CPU/RAM utilization, NVIDIA GPUs, Kubernetes clusters, and BeeGFS storage.
- Troubleshoot infrastructure, storage, networking, container, and GPU-related issues to ensure high availability and optimal cluster performance.
- Support Kubernetes-based AI workloads by assisting with deployment, resource allocation, GPU scheduling, and operational maintenance.
- Deployed and validated LLM inference on Kubernetes: a 27B-parameter Qwen model served with vLLM on an NVIDIA H200 GPU (BF16, tensor parallelism 1, 32K served context), exposing an OpenAI-compatible API for a customer application.
- Inspected the model configuration and verified architecture compatibility with the selected vLLM version before deployment; provisioned a 200 GiB RWX BeeGFS-backed PVC for model storage and validated model weights, ownership, permissions, and Kubernetes security contexts.
- Validated the inference service end to end: model discovery, health endpoint, chat completions, structured JSON output, token-level logprobs with Top-5 alternatives, per-request thinking control, and GPU memory utilization after model initialization.
- Exposed the vLLM API through a Kubernetes ClusterIP Service and NGINX Ingress with source-IP restrictions and inference-oriented proxy timeouts, and coordinated with the firewall team to enable application connectivity.
- Perform proactive system monitoring, capacity analysis, and incident resolution to improve platform reliability and performance.
- Prepare and maintain GPU utilization reports and usage documentation for customers, providing resource consumption analysis, capacity insights, and recommendations to support planning and efficient GPU allocation.
- Collaborate with application teams, researchers, and customers to optimize HPC resource utilization and provide technical guidance for AI and HPC workloads.
- Participate in infrastructure maintenance, upgrades, documentation, and operational improvements while following security and operational best practices.

01/2024 – 06/2025
HPC SUPPORT SPECIALIST, FREELANCE

- Define and document **installation procedures for new deployments** (e.g., Slurm with database replication, OpenHPC with Slurm and Warewulf, OpenPBS with H.A Corosync Peacemaker) after thorough local testing.
- Support **HPC teams**, covering system administration, InfiniBand/Ethernet network troubleshooting, job scheduling, and performance optimization.
- Perform **root cause analysis** for job failures, complex performance issues, cluster-

wide failures, and scheduling bottlenecks by analyzing system logs and identifying recurring issues for permanent fixes.
- Conduct in-depth **performance analysis and debugging** using advanced HPC diagnostic tools.
- Develop and maintain **automation scripts** for monitoring, scheduling, and cluster management tasks.
- Maintain accurate **technical documentation**, including standard operating procedures (SOPs) and runbooks, for internal teams and customer-facing knowledge bases to ensure knowledge sharing and faster issue resolution.

02/2021 – 11/2023
IT SYSTEMS ADMINISTRATOR, CYBER COFEE, TUNISIA

- **Managed multi-user computing environment** with focus on system stability, performance optimization, and user support.
- **Performed advanced hardware diagnostics** and server infrastructure repairs (CPU, storage, Networking components) .
- **Manage security protocols** and network monitoring solutions.
- **Conducted performance analysis and system optimization** to maximize resource utilization across infrastructure.
- **Maintained technical documentation** of system configurations, incident resolution procedures.

02/2016 – 12/2021
FREELANCE UI/UX DESIGNER, FRANCE

- **Designed application** interfaces with focus on usability and performance.
- **Collaborated with distributed development teams** using project management tools and shared documentation practices.
- **Utilized modern design tools** (Adobe Family) and implemented scalable front-end solutions.
- **Conducted user research** and performance analysis to optimize system interactions.

EDUCATION

2010-2013
Technical Degree in Civil Engineering - Civil Engineering School Technology

PROFESSIONAL CERTIFICATIONS

ALTAIR & HPC WORKLOAD MANAGEMENT CERTIFICATIONS

- PBS Professional Essentials Associate Specialist (link)
- PBS101 Foundation Course on PBS Pro (link)

CLOUD COMPUTING CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Certified Foundations Associate (link)
- AWS Knowledge Cloud Essentials (link)
- Introduction to Cloud Infrastructure Technologies (link)
- Introduction to Kubernetes (LFS158) (link)

AI, NETWORKING & PROJECT MANAGEMENT

- HPE Aruba Network Security Essentials (link)
- PMP® Certification Training Course (link)

- NVIDIA InfiniBand Essentials (link)
- Artificial Intelligence Fundamentals (link)

PROJECTS

- **Slurm High Availability Deployment with DRBD Replication** *(2023)*
  Deployed a highly available Slurm environment with DRBD-based database replication.
  Improved job scheduler resilience and minimized downtime.

- **OpenPBS Installation and Troubleshooting Guide** *(2024)*
  Developed step-by-step installation and troubleshooting documentation for OpenPBS.
  Simplified deployment and reduced issue resolution time for site teams.

- **OpenHPC Cluster Deployment with Warewulf, Slurm, and Lustre** *(2025)*
  Deployed an HPC cluster using OpenHPC, Warewulf, Slurm, and Lustre.
  Validated performance and provided installation procedures for future deployments.

- **LLM Inference Deployment with vLLM on NVIDIA H200 (Kubernetes)** *(2026)*
  Deployed a 27B-parameter Qwen model with vLLM on an H200 GPU, using BeeGFS-backed storage and an OpenAI-compatible API.
  Validated logprobs, JSON output, and thinking control, and exposed the service via Kubernetes Service and NGINX Ingress.

- **Ministry of Education and Economy of Azerbaijan** Phase 1 & 2 (01/2026 – 08/2026)
  Deployed, — Deployed and expanded a Dell Omnia-based HPC cluster, integrating compute, GPU, and Kubernetes worker nodes; upgraded SLURM, BeeGFS, FreeIPA, Prometheus/Grafana, and Dell OpenManage; performed system validation, benchmarking, and post-expansion health checks.

LANGUAGES

- Arabic: Native
- French: Advanced
- English: Advanced